1.6.3 ▶ ▶ ▶ ▶ Period of a simple Pendulum

A pendulum is a good example of simple harmonic motion, and the periodic motion of its swinging has been used to keep time for centuries. To find the period of its oscillation, first consider a pendulum of length l with a mass m on the end. Furthermore, consider the pendulum at a small angle θ with respect to the vertical.

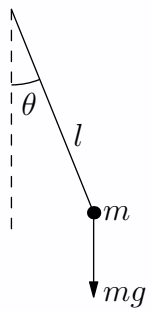

Fig. 1.16

Writing $\tau = I\alpha$ for this system about the pivot point gives

$$I\alpha = mgl\sin\theta$$

$$\alpha = \frac{mg}{ml^2}\sin\theta$$

$$\alpha = \frac{g}{l}\sin\theta \tag{1.92}$$

For small angles, $\sin\theta \approx \theta$[15]. So, we can write

$$\alpha = \frac{g}{l}\theta.$$

By the definition of SHM, we have

$$\boldsymbol{\omega} = \sqrt{\frac{g}{l}}$$

$$T = 2\pi\sqrt{\frac{l}{g}} \tag{1.93}$$

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[15] See A.2.3